

Continuous Probability Distribution

- Probability Distribution of a continuous R.V.
- Probability function of a continuous R.V.
- Probability Density function (pdf).

Let X be a ~~probable~~ continuous R.V. with probability function $f(x)$; $\forall x$.

Then, the probability function $f(x)$ is said to be probability density function (pdf) if it satisfies the following conditions:

i) $f(x) \geq 0$ $\times$ Non-negative

ii) $\int_{-\infty}^{\infty} f(x) dx = 1$ $\times$ Total prob. = 1

iii) For any particular event E such that X lies in an interval $[a, b]$. Then, ~~$P(E)$~~

$$\begin{aligned} P(E) &= P(a \leq X \leq b) \\ &= P(a < X < b) \end{aligned} \left. \vphantom{\begin{aligned} P(E) &= P(a \leq X \leq b) \\ &= P(a < X < b) \end{aligned}} \right\} \begin{array}{l} \text{Same for continuous} \\ \text{R.V.} \end{array}$$
$$= \int_a^b f(x) dx$$

Probability Distribution Function of continuous R.V.

- $\rightarrow$ Distribution Function of continuous R.V.
- $\rightarrow$ Cumulative distribution function (cdf)
 $\hookrightarrow$ distribution of cumulative prob.

Let, X be a continuous r.v. with probable pdf $f(x)$. Then the distribution function of X is denoted by $F(x)$ and is given as:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

Properties

i) $0 \leq F(x) \leq 1$

ii) $\lim_{x \rightarrow -\infty} F(x) = F(-\infty) = 0$

$$\Rightarrow F(-\infty) = P(X \leq -\infty) = 0$$

iii) $\lim_{x \rightarrow \infty} F(x) = F(+\infty) = 1$

$$\Rightarrow F(+\infty) = P(X \leq +\infty) = \int_{-\infty}^{\infty} f(x) dx = 1$$

iv) $F'(x) = \frac{d}{dx} (F(x)) = f(x)$

where, $f(x)$ is the probability density function (pdf) of X .

v) $P(a < X < b) = P(a \leq X \leq b)$

$$= P(a \leq X < b) - P(a < X \leq b)$$

$$= F(b) - F(a)$$

vi) If $x > y$, then $F(x) > F(y)$.

eg. A continuous random variable X with the following pdf.

$$f(x) = \begin{cases} k(1+x); & 2 \leq x \leq 5 \\ 0; & \text{otherwise} \end{cases}$$

i) find the value of k .

ii) Find $P(X \leq 4)$

iii) Find the distribution function of X .

→ Given; pdf $[f(x)] = k(1+x); 2 \leq x \leq 5$

i) we know,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$= k \int_2^5 k(1+x) dx = 1$$

$$= k \left[x + \frac{x^2}{2} \right]_2^5 = 1$$

$$\Rightarrow k \left[5 + \frac{25}{2} - 2 - \frac{4}{2} \right] = 1$$

$$\Rightarrow k = \frac{1}{5 + \frac{25}{2} - 4} = \frac{2}{10 + 25 - 8}$$

$$= \frac{2}{27} \quad \checkmark$$

$$10) P(X \leq 4) = \int_{-\infty}^4 f(x) dx$$

$\nwarrow$
 $P(-\infty \leq X \leq 4)$ $\leftarrow$ here have lower limit as 2

$$= \int_{-\infty}^4 \cancel{f(x) dx} \int_2^4 f(x) dx$$

$$= \int_2^4 \frac{2}{27} (1+x) dx$$

$$= \frac{2}{27} \left[x + \frac{x^2}{2} \right]_2^4$$

$$= \frac{2}{27} \left[(4-2) + \frac{16-4}{2} \right]$$

$$= \frac{2}{27} [2+6]$$

$$= \frac{16}{27}$$

iii) No, distribution function of X :

$$F(x) = P(X \leq x)$$

$$= P(2 \leq x \leq x)$$

$$= \int_2^x f(x) dx$$

$$= \frac{2}{27} \left[x + \frac{x^2}{2} \right]_2^x$$

$$= \frac{2}{27} \left[(x-2) + \frac{x^2-4}{2} \right] \quad \underline{\underline{2 \leq x \leq 5}}$$

$$F(4) = P(X \leq 4)$$

↳ we can put $x=4$ in $F(x)$ over 2

eg: A continuous R.V. X has the following

$$\text{pdf } f(x) = \begin{cases} \frac{1}{4} & ; -2 < x < 2 \\ 0 & ; \text{otherwise} \end{cases}$$

Find;

- (i) $P(X < 1)$
- (ii) $P(|X| < 1)$
- (iii) $P(2X + 3 > 5)$

Sol.

$$\text{pdf } [f(x)] = \frac{1}{4} ; -2 < x < 2$$

$$(i) P(X < 1) = \int_{-2}^1 f(x) dx$$

$$= \int_{-2}^1 \frac{1}{4} dx$$

$$= \frac{1}{4} [x]_{-2}^1$$

$$= \frac{3}{4}$$

$$(ii) P(|X| < 1) = P(-1 < X < 1)$$

$$= \int_{-1}^1 f(x) dx$$

$$= \frac{2}{4}$$

$$= \frac{1}{2}$$

$$\begin{aligned}
 \textcircled{iii} \quad P(2x+3 > 5) &= P(5 < 2x+3) \\
 &= P(5-3 < 2x) \\
 &= P(2 < 2x) \\
 &= P(1 < x) = P(x > 1) \\
 &= \int_1^2 f(x) dx \\
 &= \int_1^2 x dx \\
 &= \frac{1}{2} [x^2]_1^2 \\
 &= \frac{1}{2} (4 - 1) \\
 &= \frac{3}{2}
 \end{aligned}$$

eg; A continuous R.V. x has the following pdf;

$$\begin{aligned}
 f(x) &= 2x; & 0 \leq x \leq 1 \\
 &= 2; & 1 \leq x \leq 2 \\
 &= -2x+3; & 2 \leq x \leq 3 \\
 &= 0; & \text{otherwise}
 \end{aligned}$$

① Find a

② Find $P(X < 1.5)$

Ans:

we have:

$$f(x) = ax; \quad 0 \leq x \leq 1$$

$$= a; \quad 1 \leq x \leq 2$$

$$= -ax + 3a; \quad 2 \leq x \leq 3.$$

Now,

$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad \Rightarrow \quad \int_0^3 f(x) dx = 1.$$

$$\Rightarrow \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx = 1$$

② $P(X < 1.5)$

$$= \int_0^{1.5} f(x) dx.$$

$$= \int_0^1 f(x) dx + \int_1^{1.5} f(x) dx$$

Q. A R.V. X has the following probability distribution

X	$P(X)$
-3	0.1
-1	0.2
0	0.15
1	0.2
2	0.2
3	0.15
5	0.05
8	0.05

i) Calculate the prob. that:

ii) X is negative

$$\rightarrow P(X < 0)$$

$$iii) P(1 < X \leq 2)$$

$$iv) P(X \leq 3 / X > 0)$$

$$v) P(X > 3 / X > 0)$$

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$$P(X \leq 3 \cap X > 0) = \frac{P(X \leq 3 \cap X > 0)}{P(X > 0)}$$

Conditional probability

$$\text{Here, } P(X > 0) =$$

&

$$P(X \leq 3 \cap X > 0)$$

↙

$$= P(X=1) +$$

$$P(X=2) +$$

$$P(X=3)$$

#

-3, -1, 0, 1, 2, 3, 5, 8

Interpret.

$$\textcircled{i} \quad P(X < 0) = P(X = -3) + P(X = -2) + P(X = -1)$$

=

$$\textcircled{ii} \quad P(1 < X \leq 8) = P(X = 2) + P(X = 3) + P(X = 5) + P(X = 8)$$

=

$$\textcircled{iii} \quad P(X \leq 3 | X > 0) = \frac{P(X \leq 3 \cap X > 0)}{P(X > 0)}$$

$$= \frac{P(X = 1) + P(X = 2) + P(X = 3)}{P(X = 1) + P(X = 2) + P(X = 3) + P(X = 5) + P(X = 8)}$$

$$+ P(X = 8)$$

✓

Mathematical Expectation:

⇒ Expectation of a Random Variable.

⇒ The mean of a R.V.

⇒ The expected value of a R.V.

Let X be a R.V. Then the expectation of X is denoted by, $E(X)$ and is given as:

(i) For discrete R.V.;

$$E(X) = \sum x \cdot P(X=x), \text{ where}$$

$P(X=x)$ is the prob. of x .

(ii) For Continuous R.V.;

$$E(X) = \int x \cdot f(x) dx \text{ where } f(x) \text{ is pdf of } x.$$

range of x

eg: A Discrete probability

distribution.

R.V. X has

the following

fn: $F(x)$
or

Find the expectation of

X .
or

Find the mean of

X .
or
Find the expected value

$X=x$	$P(X=x)$
-2	0.2
-1	0.2
0	0.2
1	0.3
2	0.3

Soln.

$X = x$	$P(X = x)$	$x \cdot P(X = x)$
-2	0.1	-0.2
-1	0.1	-0.1
0	0.2	0
1	0.2	0.2
2	0.3	0.6
3	0.1	0.3
		$\sum_{x=-2}^3 x \cdot P(X=x) = 0.8$

we know,

The mean of x is;

$$E(x) = \sum_{x=-2}^3 x \cdot P(X=x)$$

$$= 0.8$$

eg; A continuous R.V. x has the following probability density function; ~~for~~

$$f(x) = \begin{cases} kx^3 & ; 0 \leq x < 1 \end{cases}$$

① Find the value of k .

② Find the mean of X . or mean of distribution

∴ $f(x)$ is pdf $\Rightarrow$ Total prob. = 1.

①

$$\int_0^1 kx^3 dx = 1$$

$$\Rightarrow \frac{k}{4} = 1 \Rightarrow \boxed{k=4}$$

$$\therefore f(x) = 4x^3; 0 < x < 1$$

② Now we know

$$\text{mean of } x \text{ i.e. } E(x) = \int_0^1 x f(x) dx$$

$$= \int_0^1 x \cdot 4x^3 dx$$

$$= 4 \cdot \left[\frac{x^5}{5} \right]_0^1$$

$$= \frac{4}{5} \text{ H}$$

Properties of Expectation:

① $E(a) = a$, where 'a' is constant

② $E(ax) = a E(x)$

③ $E(ax+b) = a E(x) + b$; where 'a' and 'b' are constants

$$\Rightarrow E(2x+5) = 2 E(x) + 5$$

④ $E(x \pm y) = E(x) \pm E(y)$

⑤ $E(xy) = E(x) \cdot E(y)$; if x and y are independent

⑥ Let, $g(x)$ be a function of x . Then,

$$E[g(x)] = \sum g(x) P(x=x); \text{ if } x \text{ is discrete R.V.}$$

$$= \int g(x) f(x) dx \quad \text{if } x \text{ is continuous R.V.}$$

eg: Suppose $g(x) = x^2$.

$$\text{Then, } E(x^2) = \sum_x x^2 P(x=x) \quad \text{if } x \text{ is discrete R.V.}$$

$$= \int x^2 f(x) dx; \text{ if } x \text{ is continuous R.V.}$$

Variance of R.V.

It measures the variation or deviation of the values of R.V. from its expected value.

(Mean), $E(x)$.

Let, x be a R.V. with expectation $E(x)$.

Then, the variance of x is denoted by;

$V(x)$ or $\text{Var}(x)$, or σ_x^2 and is calculated as;

$$V(x) = E[x - E(x)]^2$$

X is discrete r.v.;

$$E[X - E(X)]^2 = \sum [X - E(X)]^2 P(X=x)$$

on expanding, we get;

$$V(X) = E(X^2) - [E(X)]^2$$

↳ for calculation.

Note:

(i) $E(X^2) \neq [E(X)]^2$

(ii) The square root of variance is standard deviation.

i.e., $S.D. (\sigma_x) = \sqrt{V(X)}$

(iii) The value of variance and S.D. are always non-negative.

(iv) Properties of Variance:

1) $V(a) = 0$, where 'a' is constant i.e., variance of const.

2) $V(ax) = a^2 V(x)$

3) $\Rightarrow V(5x) = 25 V(x)$

4) $V(ax+bx) = a^2 V(x) + b^2 V(x)$ [where 'a' & 'b' are constants]

5) $\Rightarrow V(5+x^2) = 25 V(x)$

$$4) \cdot V(X \pm Y) = V(X) + V(Y) \pm 2 \operatorname{COV}(X, Y)$$

where, $\operatorname{COV}(X, Y)$ = Covariance of X and Y .

But if X and Y are independent;

$$V(X \pm Y) = V(X) + V(Y)$$

$$[\because \operatorname{COV}(X, Y) = 0]$$

Covariance:

It measures the variation or deviation of two R.V.s, say X and Y simultaneously.

Let, X and Y are two R.V.s with the expectations $E(X)$ and $E(Y)$ respectively. Then, the covariance of X and Y is denoted by

$$\operatorname{COV}(X, Y) \text{ or } \sigma_{xy}.$$

and is calculated as:

$$\operatorname{COV}(X, Y) = E[X - E(X)] \cdot [Y - E(Y)]$$

On expanding, we get;

$$\boxed{\operatorname{COV}(X, Y) = E(XY) - E(X)E(Y)}$$

where ; $E(XY) = \sum_x \sum_y xy \cdot \underbrace{P(X=x, Y=y)}_{\text{Joint probability distribution}} ;$

if X & Y are discrete R.V.

$$= \int_x \int_y xy f(x,y) dx dy \quad \text{if } x \text{ \& } y \text{ are continuous R.V.s.}$$

↳ joint probability distribution.

Example:

A discrete R.V. X has the following probability distribution.

$X = x$	$P(X=x)$
-2	0.1
-1	0.1
0	0.2
1	0.2
2	0.3
3	0.1

① find mean of X .

② find the variance of X .

③ Also find standard deviation.

④ Find $E(2X+5)$, $E(X+1)^2$ and $V(2X+5)$.

Sol:

$X=x$	$P(X=x)$	$x \cdot P(X=x)$	$x^2 = x^2$	$x^2 \cdot P(X=x)$
-2	0.1	-0.2	4	0.4
-1	0.1	-0.1	1	0.1
0	0.2	0	0	0
1	0.2	0.2	1	0.2
2	0.3	0.6	4	1.2
3	0.1	0.3	9	0.9
$\sum x \cdot P(X=x) = 0.8$				$\sum x^2 \cdot P(X=x) = 2.8$

① Now mean of X ;

$$E(X) = \sum_{x=-2}^3 x \cdot P(X=x) \\ = 0.8$$

② Now,

$$V(X) = E(X^2) - [E(X)]^2 \\ = 2.8 - (0.8)^2 \\ = 2.8 - 0.64 \\ = 2.16$$

$$E(X^2) = \sum_{x=-2}^3 x^2 \cdot P(X=x) = 2.8$$

③ $sd. (\sigma_x) = \sqrt{V(X)} = \sqrt{2.16}$

④ $E(2X+5) = 2 E(X) + 5 = 2 \times 0.8 + 5 = 6.6$

⑤ $E(X+1)^2 = \cancel{\sum (x+1)^2 \cdot P(X=x)} = [E(X+1)]^2 \\ = [E(X) + 1]^2 \\ V(2X+5) = 4 V(X) \\ = 4 + 2.16$

eg: A continuous r.v. x has the following pdf.

$$f(x) = xe^{-x} ; x \geq 0$$

Find mean and variance of the distribution.

Also, find SD.

Soln

Given,

$$f(x) = xe^{-x} ;$$

Now, mean is given by

$$E(x) = \int_0^{\infty} x \cdot f(x) dx$$

$$= \int_0^{\infty} x \cdot f(x) dx$$

$$= \int_0^{\infty} x \cdot xe^{-x} dx$$

$$= \int_0^{\infty} x^2 \cdot e^{-x} dx$$

ILATE

~~$$= \left[x^2 \int e^{-x} dx - \int \frac{d(x^2)}{dx} \int e^{-x} dx \right]_0^{\infty}$$~~

~~$$= \left[x^2 e^{-x} - \int 2x \cdot e^{-x} dx \right]_0^{\infty}$$~~

~~$$1(x) e^{-x}$$~~

20)

22

$$\begin{array}{c} + \\ \hline \end{array} \rightarrow -e^{-x}$$

21

$$\begin{array}{c} - \\ \hline \end{array} \rightarrow e^{-x}$$

2

$$\begin{array}{c} + \\ \hline \end{array} \rightarrow -e^{-x}$$

0

$$= \left[-x^2 e^{-x} - 2x e^{-x} - 2e^{-x} \right]_0^{\infty}$$

OR

Gamma function:

$$(i) \Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx ; \Gamma(n) = (n-1)!$$

$$(ii) \frac{\Gamma(n)}{a^n} = \int_0^{\infty} x^{n-1} e^{-ax} dx$$

$$\therefore \int_0^{\infty} x^2 e^{-x} dx = \int_0^{\infty} x^{3-1} e^{-x} dx$$

$$= \Gamma(3)$$

$$= 2!$$

$$E(x) = 2$$

Beta function:

$$(i) \beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$(ii) \beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

Also,

$$E(x^2) = \int_0^{\infty} x^2 f(x) dx$$

$$= \int_0^{\infty} x^3 \cdot e^{-x} dx$$

$$= \int_0^{\infty} x^{4-1} e^{-x} dx$$

$$= \Gamma(4)$$

$$= 3!$$

$$= 6$$

$$\begin{aligned} \text{variance } V(X) &= [E(X^2)] - [E(X)]^2 \\ &= [6 - 2^2] \\ &= 2 \end{aligned}$$

$$\therefore V(X) = 2$$

$$\text{S.D.} = \sqrt{V(X)} = \sqrt{2} \#$$

8) Same as above for;

$$f(x) = 3e^{-3x}; x > 0$$

$$\begin{aligned} \textcircled{1} E(X) &= \int_0^{\infty} x f(x) dx \\ &= 3 \int_0^{\infty} x e^{-3x} dx \end{aligned}$$

$$\text{let } \begin{cases} e^{-3x} = t \\ 3e^{-3x} dx = -dt \end{cases}$$

$$\text{let } t + 3x = t$$

$$t + 3 dx = dt$$

$$\Rightarrow \frac{dx}{dt} \frac{dt}{+3} = dx$$

$$\begin{aligned} \text{as } x \rightarrow 0; t \rightarrow 0 \\ x \rightarrow \infty; t \rightarrow \infty \end{aligned}$$

$$= 2 \int_0^{\infty} \frac{t}{(t+3)} \cdot e^{-t} \frac{dt}{3}$$

$$= \frac{1}{3} \int_0^{\infty} t \cdot e^{-t} dt$$

$$= \frac{1}{3} \Gamma(2)$$

$$= \frac{1}{3} \int_0^{\infty} t^{2-1} e^{-t} dt$$

$$= \frac{1}{3} \Gamma(2)$$

$$= \frac{2}{3}$$

$$(ii) E(x^2) = \int_0^{\infty} \underline{E(x)} x^2 f(x) dx$$

$$= \int_0^{\infty} x^2 \cdot 3e^{-3x} dx$$

$$= 3 \int_0^{\infty} x^2 e^{-3x} dx$$

$$\text{Let } 3x = t$$

$$\Rightarrow 3 dx = dt$$

$$\Rightarrow \boxed{dx = \frac{dt}{3}}$$

$$= 3 \int_0^{\infty} \left(\frac{t}{3}\right)^2 \cdot e^{-t} \frac{dt}{3}$$

$$= \frac{1}{9} \int_0^{\infty} t^2 e^{-t} dt$$

$$= \frac{1}{9} \Gamma(3)$$

$$= \frac{2}{9} \times 6$$

$$= \frac{2}{3} \text{ Hz}$$

$$\therefore, V(x) = x(x^2) - [F(x)]^2$$

$$= \frac{2}{3} - \left(\frac{1}{3}\right)^2$$

$$= \frac{2}{3} - \frac{1}{9}$$

$$= \frac{6-1}{9}$$

$$= \frac{5}{9}$$

8) for $f(x) = Ax(1-x)^2$; $0 < x < 1$.
 find, $E(x)$, $V(x)$ & S.D.

Now,

$$E(x) = \int_0^1 x f(x) dx$$

$$= \int_0^1 x \cdot Ax(1-x)^2 dx$$